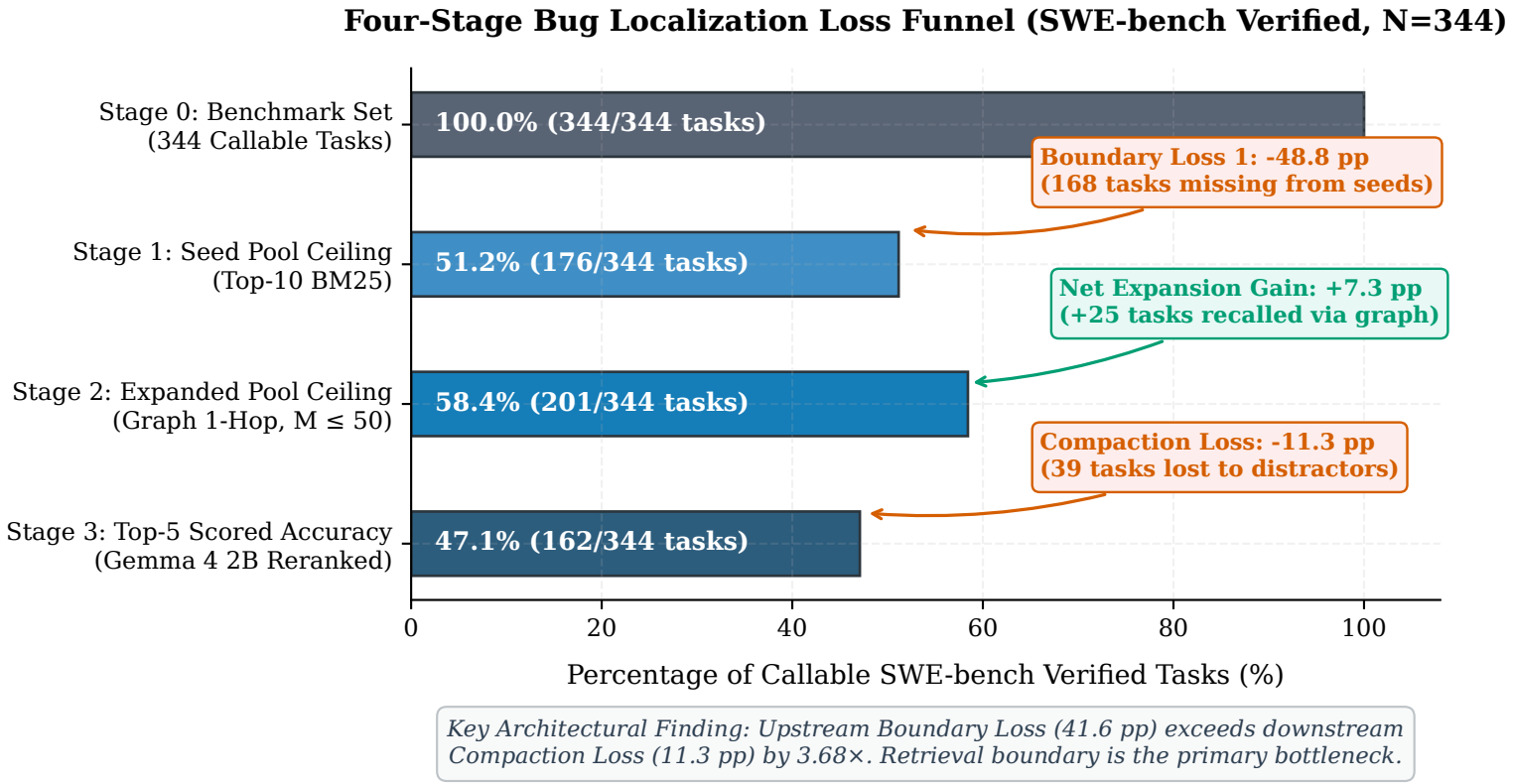
Four-Stage Bug Localization Loss Funnel (SWE-bench Verified, N=344)
Stage 0: Benchmark Set
(344 Callable Tasks)
100.0% (344/344 tasks)
Boundary Loss 1: -48.8 pp
(168 tasks missing from seeds)
Stage 1: Seed Pool Ceiling
(Top-10 BM25)
51.2% (176/344 tasks)
Net Expansion Gain: +7.3 pp
(+25 tasks recalled via graph)
Stage 2: Expanded Pool Ceiling
(Graph 1-Hop, M ≤ 50)
58.4% (201/344 tasks)
Compaction Loss: -11.3 pp
(39 tasks lost to distractors)
Stage 3: Top-5 Scored Accuracy
(Gemma 4 2B Reranked)
47.1% (162/344 tasks)
0
20
40
60
80
100
Percentage of Callable SWE-bench Verified Tasks (%)
Key Architectural Finding: Upstream Boundary Loss (41.6 pp) exceeds downstream Compaction Loss (11.3 pp) by 3.68×. Retrieval boundary is the primary bottleneck.